Inverters

The four inverters are divided in 2 groups (for each side). Each couple has got an exclusive CAN line that connect inverters to the Vehicle Control Unit. In this CAN connection are sent the following messages, as described in the datasheet [5]:

• **AMK_Actual_Values_1_XX**. The XX value change in base of the inverter taken into account: the ID are 0x283, 0x284, 0x287, 0x288 for respectively RR, FR, RL, FL inverter. In which frame message the inverter sends the signals listed below:

– **AMK_SystemReady**, that informs about the status of the inverter, it goes to 1 if the inverter is ready to work without any warning or error

– **AMK_Error**, goes to 1 if any error occurs in the inverter

– **AMK_DcOn**, informs if the Tractive system power is available in the inverter side, it goes to 1 when asserted (1 bit).

– **AMK_InverterOn**, goes to 1 when the low voltage supply of the inverter arrive in all boards (1 bit).

– **AMK_Derating**, goes to 1 when the derating function is activated

– **AMK_ActualVelocity**, informs about the actual velocity of the motor

– **AMK_TorqueCurrent**, informs about the actual torque of the motor

– **AMK_MagnetizingCurrent**, informs about the measured magnetizing current in the motor

• **AMK_Actual_Values_2_XX**. The XX value change in base of the inverte taken into account: the ID are 0x285, 0x286, 0x289, 0x290 for respectively RR FR, RL, FL inverter. In which frame message the inverte sends the signal listed below:

– **AMK_TempMotor**, return the temperature value sent by the thermocouple of the motor

– **AMK_TempInverter**, return the temperature value sent by the thermocouple of the inverter

– **AMK_ErrorInfo**, informs about possible detected temperature errors

– **AMK_TempIGBT**, return the temperature value sent by the thermocouple in the IGBT switches.

• **AMK_SetPoints_XX**, message sent by the VCU, to commands the inverters The XX value change in base of the inverter taken into account: the ID are 0x184, 0x185, 0x188, 0x189 for respectively RR, FR, RL, FL inverter. This frame is composed by the signals below:

– **AMK_InverterOn**, requests to turn on the inverter leg

– **AMK_Enable**, requests to enable the inverter to provide power to motors

– **AMK_ErrorReset**, requests a clear of all errors presents

– **AMK_DcOn**, requests to enable the Tractive System power in input of the inverters

– **AMK_TargetVelocity**, used to set the target velocity, the maximum is about 20.000 RPM

– **AMK_TorqueLimitPositiv**, used to set the maximum positive Torque to provide at the motor, this value is set by the VCU control, in order to not overcome the 80kW allowable.

– **AMK_TorqueLimitNegativ**, used to set the minimum negative Torque to provide at the motor.

AMK Actual Values 1 (0x282+n)

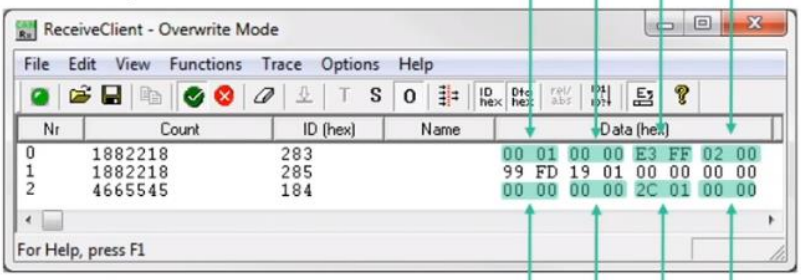
AMK_Status

AMK_ActualVelocity

AMK_TorqueCurrent

AMK_MagnetizingCurrent

CAN analyzer



AMK Actual Values 2 (0x284+n)

AMK_TempMotor

AMK_TempInverter

AMK_ErrorInfo

AMK_TempIGBT

AMK_TorqueLimitNegativ

AMK_TorqueLimitPositiv

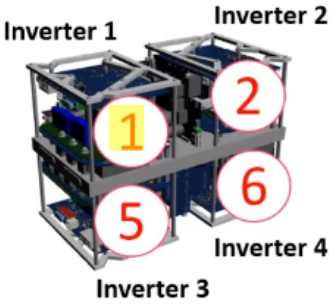
AMK_TargetVelocity

AMK_Control

AMK Setpoints 1 (0x183+n)

Calculation CAN Identifier

Node address + Basic address = CAN Identifier



Inverter	Node address	ID34091 'User list 3'		CAN Identifier	CAN Message
1	1	0x0104	0x282+01 0x284+01 0x183+01	0x283 0x285 0x184	AMK Actual Values 1 AMK Actual Values 2 AMK Setpoints 1
2	2	0x0204	0x282+02 0x284+02 0x183+02	0x284 0x286 0x185	AMK Actual Values 1 AMK Actual Values 2 AMK Setpoints 1
3	5	0x0504	0x282+05 0x284+05 0x183+05	0x287 0x289 0x188	AMK Actual Values 1 AMK Actual Values 2 AMK Setpoints 1
4	6	0x0604	0x282+06 0x284+06 0x183+06	0x288 0x290 0x189	AMK Actual Values 1 AMK Actual Values 2 AMK Setpoints 1

Beware of the same can identifier